

BATUHAN KARAMAN

✉ kbk46@cornell.edu 🌐 batuhankmkaraman.github.io

INTERESTS

LLM research & product	Reasoning LLMs, RLVR/RLHF, Alignment, VLLMs, Agents, Responsible AI
Health AI	Longitudinal data analysis, Clinical decision support, Medical Imaging

EDUCATION

Cornell University	Ithaca, NY
Ph.D. in Machine Learning (ECE)	2020 - 2026
Advisor: Mert Sabuncu	
Middle East Technical University	Ankara, Turkey
B.S. in Electrical Engineering	2015 - 2020

SKILLS

Languages	Python, R, MATLAB, C/C++
Frameworks	PyTorch, Verl, Huggingface, TensorFlow, NetworkX
Statistical Analysis	Hypothesis testing, Data visualization
Others	AWS, Kubernetes, Ray, Azure, Linux

EXPERIENCE

Amazon AGI	Sunnyvale, CA
Applied Scientist Intern	Jun 2025 - Sep 2025
Supervisors: Mohammad Ghavamzadeh, Arijit Biswas, Ruida Zhou	
– Developed a novel REINFORCE-based RLVR algorithm that achieves state-of-the-art performance on various math reasoning benchmarks using Qwen- and Llama-family models. – Analyzed the sources of instability in the REINFORCE Algorithm.	

Microsoft AI	Redmond, WA
Applied Scientist Intern	Jun 2024 - Aug 2024
Supervisors: Xia Song, Alon Benhaim, Maggie Engler (previously at Inflection AI)	
– Worked on the safety and usefulness balance of Llama- and Phi-family models through SFT and preference optimization. – Developed a novel preference optimization method, achieving a 30% reduction in model overrefusals.	

Spectral AI	Dallas, TX (Remote)
Deep Learning Scientist Intern (Part-time)	Aug 2023 - Nov 2023
– Designed a multimodal cross-attention-based model for diabetic foot ulcer healing prediction, combining multi-spectral imagery and clinical data. Achieved improved lesion localization.	

Spectral AI	Dallas, TX
Deep Learning Scientist Intern	Jun 2023 - Aug 2023
– Enhanced a multimodal convolutional model for diabetic foot ulcer healing prediction, integrating multispectral imagery and clinical data. Improved classification accuracy by 8%.	

Cornell University	New York, NY
Graduate Research Assistant	Sep 2020 - Present
Supervisors: Mert Sabuncu, Ray Razlighi (from Weill Cornell Medicine Radiology)	
– My PhD research focuses on developing computer vision and time-series methods to predict clinical outcomes from multimodal biomedical data.	

PUBLICATIONS

1. **Karaman, B.K.[†]**, Rawal, A., Shakiah S., Ghavamzadeh, M., Hong, M., Biswas, A., Zhou, R., “DISPO: Enhancing Training Efficiency and Stability in Reinforcement Learning for Large Language Model Mathematical Reasoning”, AISTATS 2026. [\[Paper\]](#)
2. **Karaman, B.K.[‡]**, Zabir, I., Benhaim, A., Chaudhary V., Sabuncu, M.R., Song, X., “POROver: Improving Safety and Reducing Overrefusal in Large Language Models with Overgeneration and Preference Optimization”, ICML 2025. [\[Paper\]](#)
3. **Karaman, B.K.^{*}**, Nguyen, M.^{*}, Kim, H.^{*}, Wang, A.Q.^{*}, Liu, F.^{*}, Sabuncu, M.R., “Knockout: A Simple Way to Handle Missing Inputs.”, TMLR, 2025. [\[Paper\]](#) [\[Code\]](#)
4. **Karaman, B.K.**, Nguyen, M., Kim, H., Sabuncu, M.R., “A Deep Survival Model for Predicting Alzheimer’s Diagnosis based on Multi-modal Longitudinal Data”, IEEE BDMA, 2025. [\[Paper\]](#)
5. **Karaman, B.K.**, Sabuncu, M.R., “Assessing the Significance of Longitudinal Data in Alzheimer’s Disease Forecasting”, AIIH 2024 (Best Paper Award). [\[Paper\]](#)
6. **Karaman, B.K.**, Dodelzon, K., Akar, G.B., Sabuncu, M.R., “Longitudinal Mammogram Risk Prediction.”, MICCAI 2024. [\[Paper\]](#) [\[Code\]](#)
7. Kim, H., **Karaman, B.K.**, Zhao, Q., Wang, A.Q., Sabuncu, M.R., “Learning-based Inference of Longitudinal Image Changes: Applications in Embryo Development, Wound Healing, and Aging Brain”, PNAS, 2024. [\[Paper\]](#)
8. Wang A.Q., **Karaman B.K.**, Kim H., Rosenthal J., Saluja R., Young S.I., Sabuncu M.R., “A Framework for Interpretability in Machine Learning for Medical Imaging”, IEEE Access, 2023. [\[Paper\]](#)
9. **Karaman B.K.**, Mormino E.C., Sabuncu M.R., “Machine learning based multi-modal prediction of future decline toward Alzheimer’s disease: An empirical study”, PLoS ONE, 2022. [\[Paper\]](#) [\[Code\]](#) (Highlighted at Cornell Chronicle on Nov 23rd, 2022. [\[Article\]](#))

INVITED TALKS & SYMPOSIUMS

1. Distinguished speaker at the 6th Global Conclave on Neurology and Neurological Disorders (NEURO Conclave 2025): “Assessing the Significance of Longitudinal Data in Alzheimer’s Disease Forecasting”.
2. Distinguished speaker at the 5th International Conference on Future of Preventive Medicine and Public Health (Future of PMPH 2025): “Longitudinal Mammogram Risk Prediction”.
3. Machine Learning in Medicine Symposium (MLIM 2022): “Machine learning based multi-modal prediction of future decline toward Alzheimer’s Disease”.

HONORS & AWARDS

- Best Paper Award, International Conference on AI in Healthcare (AIIH), 2024.
- Irwin Jacobs Scholar Fellowship, Cornell University, 2020.
- METU High Honor Award, based on graduation grades, METU, 2020.
- EEE STAR Award, given by the Electrical and Electronics Engineering Department at METU for participation in research, METU, 2019.

SERVICE

- **Reviewer:** ICLR, ICML, AISTATS, Nature, PloS, MELBA, AIIH, Journal of Alzheimer’s, etc.
- **Teaching:** Mentored multiple Cornell Master’s and Undergraduate students on their graduation projects and theses.

[†]Work done during an internship at Amazon AGI.

[‡]Work done during an internship at Microsoft AI.

^{*}Indicates equal contribution.